

Shri Aakash Padmanabhan

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Work Experience

Siemens Healthineers

Bengaluru, India

Software Engineer

July 2023–Present

- Designed and implemented robust PET-MR calibration workflows for real-time image acquisition and monitoring on MR consoles to support clinical imaging diagnostics.
- Built firmware update automation for PET systems with versioning logic, reducing manual overhead and ensuring compatibility with MR systems.
- Resolved critical memory leaks, reducing RAM usage by 40% and improving runtime performance on clinical systems.
- Developed Azure DevOps (YAML) pipelines to improve automated test coverage from 30% to 65%, accelerating regression testing by 40%.
- Handled integration of PET delivery packages into MRI systems by collaborating between two teams, reducing integration time from a few days to a few hours by automating integration steps and tests for DICOM generation and cross-modality communication functionalities of the delivery.
- Streamlined release branching and check-in workflows, successfully halving feature delivery latency to release testing environments.
- Worked with clinical and agile teams to resolve 15+ critical field defects; improved automated test coverage across unit and integration layers to prevent regressions.

Tech Stack: C#, C++, Azure DevOps, YAML, Git, Visual Studio.

Siemens Healthineers

Bengaluru, India

Software Engineering Intern

Jan 2023–July 2023

- Conducted manual testing to uncover crash and edge-case scenarios; logged and tracked bugs in Azure Boards.
- Implemented backend error reporting mechanisms integrated into UI display layers.
- Prototyped PET-MR calibration workflow using service-oriented modules in C# and C++/CLI with a component-based architecture.

Projects

AI PR Review Agent [GitHub]

2026

- Built a diff-aware AI system to analyze multi-file pull requests, combining static analysis with LLM reasoning to detect code quality issues and design risks across large codebases.
- Designed a pluggable architecture enabling extensible rule-based and ML-based analysis, supporting integration of multiple code analysis signals.
- Implemented incremental processing to analyze only changed code, reducing analysis scope and enabling scalable review workflows.
- Evaluated on real-world repositories (Flask, Django), successfully identifying high-severity issues (e.g., exception suppression, schema inconsistencies) and generating structured review outputs.

AI DevSecOps Code Modernizer [GitHub]

2026

- Developed an AI-powered modernization agent to detect security vulnerabilities (e.g., SQL injection) and generate safe refactoring patches using LLMs.
- Combined regex-based static analysis with LLM reasoning to identify multiple vulnerability patterns and produce secure transformations while preserving business logic.
- Generated developer-ready patch files (Unified Diff format), enabling safe integration into existing workflows via `git apply`.
- Validated on real-world vulnerable codebases, automatically detecting critical security flaws and producing actionable remediation outputs.

Academic Projects and Publications

Humanoid Robot – Alpha X

Jul 2022–Dec 2022

- Built a modular humanoid assistant with voice and vision-based interaction on Jetson Xavier AGX.
- Integrated wake-word detection (“Hey Alpha”) with Google Cloud STT and local NLP models for real-time voice responses (15ms latency); facial recognition (dlib) achieved 90% accuracy with 10ms latency under varied lighting.
- Enabled person-tracking and visitor management using embedded systems.

Modified SCARA for Tomato Sorting (Paper)

Jul 2022–Dec 2022

- Designed a robotic SCARA system for tomato classification using YOLOv5; achieved 97% mAP with 22ms/frame inference on Jetson Nano.
- Engineered coordinate transforms and inverse kinematics for precision robotic arm control.
- Validated full motion pipeline using MATLAB simulations.

Key Skills

Programming: C#, C++, Python, JavaScript
AI/ML: PyTorch, OpenCV, YOLOv5, dlib, scikit-learn
AI Systems: LLM-based Agents, Code Analysis, Prompt Engineering
DevOps: Azure DevOps, Git, Docker, Kubernetes (basic)
Embedded/Robotics: ROS, ROS2, Jetson AGX/Nano, Arduino, Raspberry Pi

Education

KLE Technological University	Hubballi, Karnataka
<i>B.E. Automation</i>	
<i>Minor: Computer Science and Engineering</i>	<i>2019–2023</i>
CGPA: 9.5/10 Rank: 1 Gold Medalist	

Achievements

Paper Presentation: 2nd Place – Advitya 2022 for Modified SCARA Tomato Sorting System
National Rank 1: InMovidu Tech All-India Scholarship Test (2020)